

Question: An individual faces a probability of falling sick of $p = 0.5$. Her utility function is $U(I) = 1000I - I^2$, where I is income. Assume: income when healthy = 1000; income when sick = 0. She has two insurance options:

- **Contract A:** Full insurance with a premium of 600.
- **Contract B:** Fair but partial insurance with payout q and premium $= p \times q$.

At what value of q is this individual indifferent between the two contracts? Provide a detailed explanation.